
Correct Scores, Wrong Scientific Conclusions: Auditing AI Forecasts of Chaotic Dynamics

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Abstract

1 Forecast accuracy does not by itself establish scientific forecasting ability. We test
2 five conditions that connect a score to a scientific claim: held-out generalization,
3 complete failure accounting, matched disclosure, structural identifiability, and cross-
4 system transfer. We apply these tests to k -pendulum ($k \in \{1, 2, 3\}$) and Lorenz-63
5 forecasts from language models, a time-series foundation model, learned dynamics,
6 sparse identification, and numerical integrators. Each test changes a substantive
7 conclusion. Learned-model error rises from 0.09–0.63 rad on training trajectories
8 to 0.97–1.14 rad on held-out trajectories, where linearized dynamics and nearest-
9 neighbor lookup perform better. Retaining failed forecasts lowers the reported
10 performance of models with incomplete coverage. An unmatched comparison
11 says that hiding physical constants helps; a paired intervention instead finds a
12 small +0.069 rad penalty. Two exact scale symmetries prove that the absolute
13 constants are not identifiable from angle trajectories. Increasing the Neural ODE
14 training set improves error from 1.090 to 0.861 rad but does not reach known-
15 equation integration. The pendulum ordering also fails to transfer to Lorenz-63,
16 where SINDy recovers the polynomial dynamics to numerical precision. On video-
17 tracked hardware the roles invert: the RK4 model with published constants is itself
18 imperfect, and its lead over simple baselines is gone within a second. These results
19 show that a correct leaderboard can still support incorrect scientific conclusions: an
20 unaudited score does not earn trust. We release the prompts, outputs, checkpoints,
21 seeds, and compute records needed to repeat the audit.

22 1 Introduction

23 Scientific claims require more than low forecast error. A leaderboard states what was measured, not
24 what inference it supports, and that gap is where trust in an evaluation is won or lost: a result may
25 depend on training trajectories, dropped failures, unmatched inputs, an unidentifiable target, or one
26 favorable system. We study the evaluation protocol itself, not only the models it ranks. Given state $\mathbf{s}_t$,
27 the task is to predict $\mathbf{s}_{t+T}$. The k -link pendulum is integrable at $k=1$ and chaotic for $k \geq 2$, has high-
28 precision ground truth, and permits controlled changes to physical metadata. Lorenz-63 provides a
29 second system with a different observation model and hypothesis class; a hardware replication of the
30 audit (§5.5) scores the reference model itself as the imperfect one.

31 Our contribution is a claim audit for dynamical-system forecasts. It tests held-out generalization,
32 counts every failed forecast, records the information available to each method, changes disclosure
33 on identical trajectories, checks identifiability before scoring parameter recovery, and tests whether
34 conclusions transfer to a second system. The five are indexed to the claims, not drawn from a
35 taxonomy: each tests one inference step between a score and a conclusion (sample to population,
36 answered cells to full task, correlation to attribution, accuracy to parameter recovery, one system
37 to a family claim), and claims we do not make would need other checks, such as calibration for

Table 1: Conclusions before and after the claim audit. Each row changes one part of the evaluation while preserving the relevant model outputs or comparison set.

Test	Unaudited conclusion	Audited conclusion
Failures	Answered forecasts represent model performance.	Incomplete coverage lowers reliability-adjusted performance.
Generalization	Learned dynamics forecast accurately.	At this budget, held-out error is up to $11\times$ in-sample, and simple baselines perform better.
Disclosure	Withholding constants improves forecasts.	Matched, the effect is small and model-dependent ($+0.069$ rad pooled).
Identifiability	Absolute constants can be recovered from angles.	Two exact symmetries make absolute recovery impossible.
Transfer	The pendulum order ranks model families.	Lorenz-63 produces a different order.

probabilistic forecasts. Table 1 states the conclusion before and after each test. The central result is empirical and direct: all five tests alter or reject a conclusion supported by the unaudited comparison, and these are conclusions the field draws: zero-shot forecasters are routinely scored without failure accounting or matched controls [Gruver et al., 2023, Zhang and Gilpin, 2025].

2 Related work

Learned dynamics. Neural ODEs learn a continuous-time vector field from data [Chen et al., 2018]. Hamiltonian Neural Networks impose an energy-conserving inductive bias through a learned Hamiltonian [Greydanus et al., 2019]; Lagrangian Neural Networks enforce the Euler–Lagrange equations through second-order autodifferentiation [Cranmer et al., 2020]. Chaotic systems such as the Lorenz attractor and the double pendulum are standard testbeds [Gilpin, 2021]. We evaluate all three against general-purpose models under matched conditions.

Foundation models as forecasters. LLMs can act as zero-shot time-series forecasters when series are serialized as text [Gruver et al., 2023], and time-series foundation models such as Chronos [Ansari et al., 2024], TimesFM [Das et al., 2024], and Moirai [Woo et al., 2024] are trained for general forecasting. Zhang and Gilpin [2025] study zero-shot forecasting of chaotic systems with such models. We evaluate these forecasters on the same grid as physics-informed and language models.

LLMs for physical and mathematical reasoning. LLMs perform well on mathematical problem solving [Lewkowycz et al., 2022] and, with chain-of-thought prompting, on multi-step reasoning [Wei et al., 2022]; recent benchmarks probe physics reasoning in LLMs [Xu et al., 2025]. Here the output is a numerical state at a specified time in a chaotic trajectory.

Data-driven identification and abstention. SINDy fits sparse governing equations from measured trajectories [Brunton et al., 2016]; nonlinear vector autoregression provides a compact reservoir-computing formulation [Gauthier et al., 2021]. Structural identifiability asks which parameter combinations can be recovered from observed outputs, including under parameter symmetries [Borgqvist et al., 2024]. Selective prediction ordinarily reports risk together with coverage [Geifman and El-Yaniv, 2017]. Here failed cells remain in the main score because models are compared at fixed task coverage.

3 Benchmark design

Choice of systems. We use the k -link pendulum to vary the number of coupled links while retaining a high-precision numerical solution. Its constants can also be hidden without changing the underlying trajectory, which permits the matched test in §5.2. The smaller Lorenz-63 experiment repeats the local-model comparison on a second chaotic system; the LLM and disclosure experiments were not rerun there.

Table 2: Inputs available to each family. State is at $t=0$; history is pre- $t=0$. Chronos-2 is history-conditioned; learned hidden cells withhold hidden constants.

Family	State	History	Constants	Equations
Numerical (RK4/Euler/sympl.)	✓	—	✓ (all regimes)	✓ (exact)
Learned (NODE/HNN/LNN)	✓	—	✓ base; withheld*	— (learned)
Time-series (Chronos-2)	—	✓ (10 s context)	—	—
LLMs (kimi/grok/deepseek)	✓	—	✓ except <code>changed_hidden</code>	—

71 3.1 Task and systems

72 Given the state $\mathbf{s}_0 = (\theta_1, \dots, \theta_k, \omega_1, \dots, \omega_k) \in \mathbb{R}^{2k}$ of a k -link pendulum at $t=0$, the task is to
 73 predict $\mathbf{s}_T$ at horizon T . Ground-truth trajectories integrate the full nonlinear Lagrangian equations of
 74 motion with fixed-step RK4 at $\Delta t=10^{-3}$ s, recording states every 0.01 s plus a 10 s pre- $t=0$ context
 75 window (used only by the time-series models). Initial conditions are sampled $\theta_i \sim \mathcal{U}(-3, 3)$ rad,
 76 $\omega_i \sim \mathcal{U}(-1, 1)$ rad/s with a fixed seed. Because the labels are produced by RK4 at this step, the
 77 RK4 baseline reported later (the same integrator at a $10\times$ coarser $\Delta t=10^{-2}$) is a numerical reference
 78 by construction, not an independent competitor. Euler and the symplectic integrator are distinct
 79 known-physics methods. Inputs available to each family are stated in §3.3.

80 3.2 Regimes and grid

81 Three regimes test generalization to non-standard constants (Table 9). In `changed_hidden` the
 82 constants (g, L, m, d) , with damping d , are not disclosed to the model. This is a matched metadata-
 83 disclosure test; a single state contains no observed dynamics from which to identify the constants. An
 84 evaluation cell is a tuple (model, k , regime, horizon), with horizons $\{0.01, 1, 10, 60\}$ s. The full LLM
 85 grid also varies input modality and prompting; every comparison here uses the coordinate modality
 86 with direct prompting (the other axes are analyzed in Appendix A).

87 **Out-of-sample protocol.** Learned dynamics models are never scored on their training data. We
 88 generate a shared held-out set (a separate seed from the training trajectories) of 180 initial conditions
 89 (60 per regime, spanning $k \in \{1, 2, 3\}$) and evaluate every family on the identical trajectories, keyed
 90 by `movement_id`. The cross-family comparison (§5.1) uses horizons $\{1, 10\}$ s, where chaos has
 91 begun but error has not saturated (§3.4), giving $180 \times 2 = 360$ cells per model; in-sample error
 92 appears only once, to quantify the size of the generalization gap (§5.1).

93 3.3 Inputs and grouping

94 Table 2 states what each family receives. Chronos is a time-series model, so we evaluate it in its
 95 standard form: a history-conditioned forecaster with a 10 s pre- $t=0$ context window. The LLMs and
 96 learned dynamics models are state-conditioned from $t=0$. For learned models, hidden-regime cells
 97 are equalized by withholding hidden constants and using normal-regime prior constants. A separate
 98 asymmetry runs in the opposite direction: the LLMs are large hosted models, whereas the learned
 99 models are small (256-wide, 3-layer MLPs trained on a single 8 GB GPU; Appendix B). The behavior
 100 of larger physics-informed networks is outside this study’s scope. We separate known-physics
 101 methods (RK4, Euler, symplectic) from black-box methods (LLMs, Chronos, learned dynamics); the
 102 former are numerical references.

103 3.4 Metrics

104 The primary metric is mean absolute angle error (rad), wrapped to $[-\pi, \pi]$ and averaged over links
 105 and trajectories. We also track Cartesian bob error, angular-velocity sign match, and energy drift.

106 **Reliability-adjusted error.** A cell is answered only if the model returns a parseable, finite prediction.
 107 Reporting error over answered cells alone would reward abstention on hard cells. We therefore assign
 108 unanswered cells the random-guess baseline $\pi/2$ (the expected wrapped-angle error of a uniform
 109 guess); under the strict upper bound π only the incomplete-coverage models move (§5.1). Answering

110 is part of the score: the task is fixed-coverage forecasting, not selective prediction. Angles alone are
 111 scored, since wrapping gives answers and failures one bounded scale that unbounded velocities lack.

112 **Verification record.** For each evaluation cell we retain the trajectory identifier, model and de-
 113 ployment name, prompt, raw response, parser outcome, prediction, prompt and completion token
 114 counts, and score. The released configuration fixes the data-generation solver, held-out seed, horizons,
 115 prompts, timeout, and temperature. Two standards apply here. Local methods are reproducible:
 116 their checkpoints and seeds regenerate the predictions exactly. Hosted models are not, because
 117 a deployment can change or be withdrawn between evaluations, so for them the record is what
 118 remains. We therefore release the per-cell responses rather than summary scores alone, which keeps
 119 the reported API evaluation auditable after the deployments it measured are gone.

120 **Predictability horizon.** For chaotic systems, pointwise error past the Lyapunov time saturates to a
 121 system-dependent constant for every model, so long-horizon error rankings carry little information.
 122 We define the predictability horizon as the longest horizon a model tracks contiguously below a
 123 threshold τ , starting from $t=0$: we walk horizons in increasing order and stop at the first that
 124 exceeds τ . The contiguous rule is used because, after divergence, a later horizon can fall below τ by
 125 angle-wrap coincidence. We sweep τ .

126 **Disclosure gap.** We define $\Delta_{\text{disc}} = \text{err}(\text{hidden}) - \text{err}(\text{disclosed})$. Positive values indicate a penalty
 127 for withholding constants. Because regime contrasts confound disclosure with trajectory difficulty,
 128 we also run a matched experiment (§5.2) that toggles disclosure on identical trajectories.

129 4 Methods

130 **LLMs.** We evaluate `kimi-k2.6`, `grok-4-1-fast-reasoning`, and `deepseek-v4-pro` through
 131 an OpenAI-compatible API (hosted deployments, June 2026), with a structured prompt (system
 132 description, disclosed constants where applicable, initial state) and a JSON answer schema (Ap-
 133 pendix A). Each cell uses one temperature-0 call with a uniform token and timeout budget; no
 134 best-of- N aggregation is used. The LLM evaluation (Parts A and B, 3,960 cells) records prompt
 135 and completion token counts and has an estimated nominal API cost of \$110.38 under the pricing
 136 assumptions in the released analysis script.

137 **Time-series foundation models.** We run Chronos-2 [Ansari et al., 2024] locally on a single GPU
 138 in its standard history-conditioned form: the 10 s pre- $t=0$ context window of all $2k$ state channels,
 139 resampled to at most 512 points, forecasts to the target horizon (prediction length ≤ 64 steps). The
 140 comparison uses `chronos-2` in univariate and multivariate modes; the latter forecasts all $2k$ channels
 141 jointly. The paired mode comparison uses 20 trajectories per (k, regime) cell.

142 **Learned dynamics.** Neural ODE, HNN, and LNN share a 256-wide, 3-layer tanh MLP and
 143 integrate at inference with `scipy`. We compare two objectives: derivative matching (MSE on finite-
 144 difference $\dot{s}$) and rollout loss (MSE on the state after w Euler steps, with gradients through the unroll).
 145 We also use mixed-window rollout, which trains on several window lengths jointly (Neural ODE:
 146 $w \in \{10, 50\}$; HNN: $w \in \{5, 20\}$). The LNN’s loss requires second-order automatic differentiation;
 147 the LNN trains at full scale (67k samples, hidden 256) on a CUDA GPU. The primary models are
 148 trained on nine trajectories, one for each (k, regime) cell (Appendix B). A nested budget study
 149 trains fresh Neural ODE and HNN checkpoints with $\{1, 5, 20\}$ trajectories per eligible cell and
 150 seeds $\{101, 102, 103\}$. The Neural ODE uses all nine cells; because damping violates Hamiltonian
 151 conservation, the HNN budget study uses only the three undamped normal-regime cells. Every
 152 checkpoint is evaluated on the same two held-out seeds.

153 **Integrators.** Euler, RK4, and Störmer–Verlet (symplectic) at $\Delta t=0.01$ s using the exact equations
 154 of motion; their error is from discretization alone. RK4 generates the data and is included as a
 155 numerical reference, not an independent competitor.

156 **Baselines.** Four reference baselines use the same grid: persistence ($\theta_T=\theta_0$), constant velocity
 157 ($\theta_T=\theta_0+\omega_0 T$), a small-angle linearized model, and a nearest-neighbor analog forecast from the

158 matching (k, regime) training trajectory. The nearest-neighbor baseline uses the same data as the
 159 learned models.

160 On both held-out seeds (equal-weighted; §5.3) we also fit SINDy with a degree-two polynomial library
 161 and STLSQ, and nonlinear vector autoregression (NVAR) with two delays, degree-two features, and
 162 ridge regression. Divergent rollouts are retained as failures and receive the same $\pi/2$ penalty.

163 **Lorenz replication.** A secondary Lorenz–63 test compares numerical references, persistence,
 164 SINDy, NVAR, a Neural ODE trained on 20 single-seed trajectories, and zero-shot Chronos-2 on
 165 five five-trajectory held-out seeds. We report training-normalized coordinate MAE at $\{0.1, 0.5, 1.0\}$ s
 166 (full configuration in Appendix C). HNN and LNN are omitted: Lorenz–63 is dissipative and
 167 non-Hamiltonian.

168 5 Results

169 5.1 Failure accounting and held-out generalization

170 The shared held-out grid contains 360 cells: 180 trajectories at horizons $\{1, 10\}$ s. Every model
 171 is evaluated on that grid. The Answered column in Table 3 gives the fraction of cells with valid
 172 predictions. Confidence intervals are 95% paired bootstrap over trajectories ($B=10^4$); for LLMs they
 173 exclude per-call sampling variation.

174 Table 3 ranks the models on the pendulum under the access conditions in §3.3. The ranking is
 175 conditional on those inputs, this training budget, and this system; Section 5.4 tests whether it transfers.
 176 The numerical integrators sit at the top as references (RK4 0.059 at a finer step, symplectic 0.205,
 177 Euler 0.405), and every learned-dynamics model lands closer to the $\pi/2$ random baseline than to
 178 the RK4 and symplectic references. The lowest-error black-box entry is the frontier LLM kimi
 179 (0.538 rad, 99% answered), and its marginal interval excludes the nearest baseline’s. That is a
 180 statement about one recorded June 2026 deployment under these evaluation conditions, not about
 181 model families; the Lorenz test rejects the family reading.

182 Under the equalized hidden-regime run, nearest-neighbor lookup (0.725) and the linearized pendu-
 183 lum (0.855) beat every learned model, and persistence (1.004) beats all but HNN-rollout (0.972),
 184 though those two intervals overlap. The stable learned models score 0.09–0.63 rad on their training
 185 trajectories against 0.97–1.14 held out (Table 4), a $1.7\text{--}11\times$ gap widest for the in-sample winner, the
 186 mixed-window Neural ODE.

187 Two further data-driven methods appear only on the two-seed replication subset, so we report them in
 188 Table 5 rather than in the main comparison: sparse identification (SINDy, 0.883 rad on that subset)
 189 falls between the Chronos family and the learned models, and nonlinear vector autoregression (NVAR,
 190 1.248) is among the weakest data-driven methods, near constant velocity.

191 **Effect of missing predictions.** grok answers 82% of the grid and lnn 94%: their ranks fall when
 192 missing cells receive $\pi/2$, and a mean over answered cells alone would flatter both. Under the strict
 193 π bound only these two move: the LNN stays last, while grok falls eight places. The cross-family
 194 ordering is stable to the imputation constant; the rank of a model with incomplete coverage is not—the
 195 reason to score coverage rather than report it separately.

196 5.2 Matched disclosure and identifiability

197 On the shared held-out set the hidden regime has lower error than the disclosed regime (-0.390 rad,
 198 $p_{\text{Holm}} < 10^{-3}$, unpaired) and than the normal regime (-0.524). This looks favorable to withholding
 199 constants, but the regimes use different trajectories, and the hidden ones are slower and less divergent,
 200 so the comparison cannot speak to parameter recovery. When we instead toggle disclosure on
 201 identical trajectories for the three LLMs (900 paired cells on a separate matched grid; Table 8), the
 202 pooled effect reverses: hiding the constants increases error ($\Delta_{\text{disc}} = +0.069$ rad, $p_{\text{Holm}} = 0.034$), with
 203 a strong penalty at $k=1$ and a small reverse effect at $k=3$. The unadjusted hidden-regime advantage
 204 is thus a difficulty confound.

205 The disclosure test also has a structural limit. The angular equations are unchanged by $(m, d) \mapsto$
 206 (am, ad) and by $(L, g, d) \mapsto (bL, bg, b^2d)$, so for $k=1$ only g/L and $d/(mL^2)$ are identifiable from

Table 3: Out-of-sample cross-family comparison on the shared held-out set (360 cells, horizons $\{1, 10\}$ s). Reliability-adjusted mean angle error (rad): unanswered or missing cells scored at $\pi/2$; 95% bootstrap CI. Inputs differ by family (Table 2); the ranking is conditional on them, this budget, and this system. RK4 generates the data and is included as a numerical reference. Configurations first evaluated in this work are marked $\dagger$; simple baselines are marked $\ddagger$.

#	Model	Err	95% CI	Answered	Family
–	rk4 (reference)	0.059	[0.032, 0.089]	100%	numerical
1	symplectic	0.205	[0.163, 0.251]	100%	numerical
2	euler	0.405	[0.344, 0.470]	100%	numerical
3	kimi-k2.6	0.538	[0.475, 0.602]	99%	LLM
4	chronos2-multi $\dagger$	0.684	[0.617, 0.752]	100%	time-series
5	chronos2-uni $\dagger$	0.697	[0.631, 0.766]	100%	time-series
6	nearest-neighbor $\ddagger$	0.725	[0.657, 0.797]	100%	baseline
7	linearized $\ddagger$	0.855	[0.774, 0.939]	100%	baseline
8	grok-4-1-fast-reasoning	0.881	[0.805, 0.957]	82%	LLM
9	deepseek-v4-pro	0.952	[0.881, 1.027]	100%	LLM
10	hnn-rollout	0.972	[0.898, 1.047]	100%	learned
11	persistence $\ddagger$	1.004	[0.934, 1.078]	100%	baseline
12	neural-ode-rollout-mixed $\dagger$	1.036	[0.967, 1.104]	100%	learned
13	hnn-rollout-mixed $\dagger$	1.066	[0.990, 1.143]	100%	learned
14	neural-ode	1.110	[1.039, 1.181]	100%	learned
15	neural-ode-rollout	1.121	[1.054, 1.190]	100%	learned
16	hnn	1.138	[1.059, 1.218]	100%	learned
17	constant-velocity $\ddagger$	1.319	[1.241, 1.396]	100%	baseline
18	lnn $\dagger$	1.356	[1.282, 1.430]	94%	learned

an angle trajectory; absolute g , L , m , and d cannot be scored as independently recoverable. Asked to name the constants from a single state, the LLMs mostly return Earth-standard defaults, which is evidence of a prior, not of recovery (transcripts in the released records). A released check verifies both invariances to below 10^{-13} .

5.3 Replication across seeds and training budgets

We reran the local pendulum subset on an independent held-out seed with five trajectories per (k, regime) cell; the original seed has 20. Table 5 reports each seed separately and gives their unweighted mean. The selected ordering is unchanged. The numerical references have the lowest errors, Chronos-2 precedes the learned models, and HNN-rollout has the lowest error among the selected learned models. Two seeds do not estimate seed-level uncertainty. The LNN needs a note: its low coverage was a hardware artifact, not model behavior. Its inference solves a Hessian at each step, which exhausted the 8 GB GPU on the longest rollouts and returned those cells as failures. Re-running the LNN on CPU is the same evaluation with the same forecasts; it only recovers the dropped cells, raising primary-seed coverage from 66% to 94%. We use this CPU evaluation for the LNN on both seeds (the second answers all 90 cells, equal-seed rate 97%), so its numbers are produced identically and are comparable across seeds. The cells that still fail are genuine autograd-ODE divergences, scored at $\pi/2$ like any other failure, and the mean stays high at 1.366 rad.

Training-budget check. With three training initializations and the two held-out seeds, Neural ODE error decreases from 1.090 at one trajectory per cell to 0.968 at five and 0.861 at twenty (Table 10). HNN error on the normal regime falls from 1.269 to 0.974 and then remains 0.981. The HNN and Neural ODE rows cover different regimes and are not an architecture comparison. More data helps the Neural ODE monotonically, the HNN gains plateau by 15 trajectories, and neither closes the gap to known-equation integration.

5.4 Cross-system test: Lorenz–63

We test whether the pendulum conclusions transfer to Lorenz–63, a chaotic system with a polynomial vector field. They do not. Both the identifiability result and the model ordering change.

The identifiability verdict flips. On the pendulum the physical constants are not recoverable from an angle trajectory (§5.2); on Lorenz, SINDy recovers the governing equations themselves, not merely accurate forecasts. The refit selects exactly the true sparsity pattern and returns $(\hat{\sigma}, \hat{\rho}, \hat{\beta}) = (10, 28, 8/3)$ to coefficient error 2.2×10^{-13} (Appendix C); forecast error stays below 2×10^{-5} in every seed. The difference is the setting, not the audit: Lorenz–63 lies inside SINDy’s hypothesis class and is observed in full state, so its parameters are identifiable, whereas the pendulum’s absolute constants are not identifiable from angles under any method. Identifiability is a property of the observation model and the hypothesis class, and the check makes it explicit before recovery is scored.

The model ordering also changes. Trained on one seed and evaluated on five held-out seeds (25 trajectories), a Lorenz Neural ODE reaches normalized MAE 0.014 ± 0.008 , 0.079 ± 0.030 , and 0.277 ± 0.063 at horizons 0.1, 0.5, and 1.0 s (mean $\pm$ standard error over seeds), below both Chronos-2 modes at every horizon (multivariate 0.101 ± 0.018 , 0.691 ± 0.129 , 0.771 ± 0.097 ; univariate 0.126 ± 0.027 , 0.700 ± 0.102 , 0.732 ± 0.085). NVAR is close at 0.1 s (0.012) but drifts to 0.429 by 1.0 s, a persistence baseline trails (0.900 at 1 s), and divergent rollouts are scored at $\pi/2$ as on the pendulum (Table 5). This is the reverse of the pendulum, where the trained models lost to trivial baselines: here a system-trained model beats the zero-shot foundation model. As on the pendulum the families see different inputs (SINDy the library, the Neural ODE target-system training, Chronos a history window), so this is not an architecture race; the cross-system result rejects the pendulum ranking as a general ordering of model families.

The reversal does not rest on a favorable seed. The Neural ODE beats both Chronos-2 modes in every one of the 15 seed-by-horizon comparisons, and SINDy stays below 1.7×10^{-5} in every cell. The identifiability half of the contrast is not a sampling claim: the pendulum symmetries are exact and verified to below 10^{-13} , and Lorenz–63 lies in SINDy’s library by construction.

The Lorenz test includes the local methods for which the comparison is well defined. It excludes the API LLM grid and the Hamiltonian and Lagrangian networks; a Hamiltonian model does not match the dissipative Lorenz flow. Those omissions limit what the test can establish, but not what it can refute: the pendulum result is only interesting if it generalizes, and one system on which the ordering and the identifiability verdict both invert is enough to withdraw that generalization. We claim no more than that, and in particular the Lorenz ordering is not offered as the correct one.

5.5 Real data: the reference model becomes the imperfect one

On real hardware the RK4 integrator is no longer ground truth but the imperfect model. We use an open video-tracked double pendulum [Myers et al., 2020] (three free-drop trials, 500 Hz, published point-mass constants, $\sim 0.35^\circ$ median tracking noise); Trial 1 serves only for calibration (nearest-neighbor database, SINDy fit, one-parameter damping fit), and Trials 2–3 supply 120 held-out states scored as before at $\{0.1, 0.5, 1.0\}$ s (Table 15, Appendix F).

RK4 with the published constants leads at 0.1 s (0.084 rad, about $14\times$ the tracking noise) but reaches 1.25 rad by 1.0 s, indistinguishable from persistence (1.23; paired difference $+0.02$, CI $[-0.12, +0.16]$); unmodeled friction and the point-mass simplification consume its advantage within a second. The fitted damping trades horizons rather than dominating (0.89 vs. 0.82 at 0.5 s, 1.15 vs. 1.25 at 1.0 s; the 1.0 s gain is significant, CI $[-0.20, -0.00]$). SINDy, exact on Lorenz, sits at baseline level here (1.54); the hardware’s dynamics are not in its library. Chronos-2 answers 89% of cells, since states shortly after the drop lack the 5 s context used here (Appendix F); failure accounting scores them at $\pi/2$. Every check transfers, and the ordering differs from both synthetic orderings.

5.6 Diagnostics

Behavior by horizon. Out-of-sample error by horizon appears in Table 13 (Appendix E). At 1 s the ordering follows the main comparison: the integrators are near-exact, kimi (0.457) has the lowest black-box error, and the learned models are at 0.86–1.25. At 10 s all models are in the chaotic regime, and the ordering is stable: kimi (0.619) remains lowest among black-box methods while the learned models reach 1.09–1.46. The mixed-window Neural ODE is intermediate at both horizons (0.951/1.120). Figure 1 shows the growth curves over the $\{0.01, 1, 10, 60\}$ s ladder: the numerical integrators stay near zero until ~ 10 s, whereas every black-box family saturates near the chaotic noise floor by 1 s. The $\{1, 10\}$ s window used for the comparison is the range over which the families remain separable.

Breakdown by k and regime. Tables 6 and 7 (Appendix D) report out-of-sample error by number of links and by regime. Error increases with k for every model except the mixed-window Neural ODE, which dips at $k=3$ (for `kimi`, $0.11 \rightarrow 0.55 \rightarrow 0.95$; the integrators are near-exact at $k=1$ and degrade as chaos sets in), and the cross-family ordering holds within each k . The `changed_hidden` regime has the lowest error for most models (for `kimi`, 0.46 hidden vs. 0.64 normal). For the learned models, withholding the constants reduces but does not remove the difficulty effect: hidden-regime errors are now 0.71–1.19; the earlier values favor models given the true constants. The low-gravity hidden constants ($g=1.62$) produce slower, less-divergent motion over a fixed horizon, so the unadjusted hidden-regime advantage is a difficulty confound (§5.2). A cluster bootstrap over complete held-out trajectories changes no confidence interval by more than ~ 0.03 rad and leaves the ranking unchanged.

Training-objective check. Table 11 reports the effect of the loss with architecture fixed. Under the equalized hidden-regime run, rollout loss is not uniformly better: mixed-window rollout is lower for the Neural ODE, while single- k rollout is lower for the HNN. Full-scale CUDA training of the LNN gives the highest learned-model error (1.356), with 6% of cells diverging or exceeding the rollout timeout after memory-safe recovery. We make no architecture-wide claim about LNNs; under our implementation the LNN is numerically unstable and generalizes poorly out of sample.

Physical-fidelity check. Hamiltonian and Lagrangian networks target structure preservation, so we evaluate energy drift on the undamped (`normal`) regime, where true energy is constant. Table 12 reports mean absolute drift over 10 s, normalized by $E^* = g \sum_i m_i L_i$, and the fraction of bounded rollouts. The learned models do not match numerical integrators on this axis: the lowest-drift learned model is 0.779, compared with 0.066 for the symplectic integrator and 0.494 for Euler. Symplectic rollout of the HNN leaves drift nearly unchanged, so the integrator does not account for the HNN’s drift. At the derivative-matching objective, HNN drift is comparable to Neural ODE drift (4.27 vs. 4.35); the LNN is unstable, with 27% bounded rollouts. These results use the nine-trajectory budget of the main comparison.

Multivariate time-series check. On 720 paired samples, Chronos-2 multivariate and univariate errors are indistinguishable: 0.543 (95% CI [0.498, 0.589]) versus 0.549 ([0.505, 0.593]). No effect remains after Holm correction, so the larger study rejects the apparent per- k trend in the primary grid.

6 Discussion

The audit changes five conclusions the raw comparison supports (Table 1), and the changes are to the scientific claim, not to presentation. More data improves the Neural ODE from 1.090 to 0.861 rad without closing the gap to known-equation integration; HNN improvement plateaus. On Lorenz, SINDy is nearly exact because the vector field lies in its library, while a system-trained Neural ODE beats zero-shot Chronos-2. Performance therefore depends on system, observations, hypothesis class, data, and access—conditions a scalar leaderboard does not encode.

7 Limitations

The full LLM grid uses one held-out seed (the local replication adds a second, with fewer trajectories per cell); the budget study uses three initializations and two seeds; inputs differ by method (Table 2). The Lorenz test excludes API LLMs, HNN, and LNN; the hardware check has three trials of one apparatus with filtered velocities; SINDy needs the true polynomial degree in its library. Predictions are single endpoint states, conservation tests cover only the undamped regime, and we cannot rule out pendulum-like trajectories in the foundation models’ training corpora.

8 Conclusion

Correct scores can support wrong scientific conclusions: generalization, failure accounting, matched disclosure, identifiability, and cross-system transfer each change a claim the raw leaderboard supports, so benchmarks should report those checks, budgets, and access with the result. The synthetic systems give every check exact ground truth, which is what validates the audit itself; the hardware check then runs it where the reference model is imperfect—the regime real evaluations operate in.

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A Prompt templates

The prompts used for every LLM evaluation cell are reproduced below verbatim.

380 **System prompt (direct / no-CoT).**

381 You are a careful physicist predicting the future state of a k -pendulum (a chain of
382 k point masses on massless rods, hinged in a planar chain, swinging under gravity
383 with viscous damping). Respond with ONLY a single JSON object on one line, no prose,
384 no code fences. Schema:
385 {"theta": [...k floats..], "omega": [...k floats..]}
386 theta_i is in radians (link i measured from straight-down,
387 positive counter-clockwise)
388 and omega_i is angular velocity in rad/s.

389 **System prompt (chain-of-thought).** Same preamble, then: "Reason step-by-step about the dynamics,
390 then output your final answer. Wrap the final answer in <answer>...</answer> tags containing ONLY a
391 single JSON object: {"theta": [...k floats..], "omega": [...k floats..]}."

392 **User prompt (coordinate modality).**

393 Number of links $k = \{k\}$
394 Physical constants:
395 g (gravity, m/s^2): {g}
396 L (rod lengths, m): {L}
397 m (point masses, kg): {m}
398 damping (viscous coefficient on omega): {damping}
399 Initial state at $t=0$:
400 theta_0 (rad): {theta0}
401 omega_0 (rad/s): {omega0}
402 Predict the state at $t = \{T\}$ seconds.
403 Return theta and omega as length- $\{k\}$ JSON arrays.

404 **Hidden-constants regime.** The constants block above is replaced by:

405 Physical constants: HIDDEN. The numerical values of gravity, rod lengths, point
406 masses, and damping are NOT disclosed. You must infer dynamics from the initial
407 conditions alone.

408 **Parsing and scoring.** Each response is stripped of code fences and (for CoT) of everything outside the
409 <answer>...</answer> tags; the first {...} object is JSON-parsed and validated to contain length- k theta
410 and omega arrays. A response that fails any check is counted as unanswered and scored at $\pi/2$.

411 **Prompting, modality, and repeatability axes.** The released grid also varies prompting and input
412 modality; from the saved records (trajectory-level paired bootstrap throughout): (i) Chain-of-thought does not
413 change the model ordering, and for the best-scoring LLM it is significantly *worse* under reliability-adjusted
414 scoring (0.659 vs. 0.538 rad; paired difference +0.121, CI [+0.061, +0.183]): reasoning consumes the token
415 budget, unanswered cells rise from 6 to 50 of 360, and failure accounting—not reasoning quality—drives
416 the gap. For the other two models CoT is indistinguishable (−0.038, CI [−0.102, +0.025]; −0.041, CI
417 [−0.107, +0.021]). (ii) On the one-trajectory-per-cell modality grid, replacing coordinates with a rendered
418 image never helps (significantly worse for one model, +0.210, CI [+0.033, +0.392]), adding an image to
419 coordinates changes nothing measurable, and the model without vision support fails every image cell, which
420 fixed-coverage scoring records at $\pi/2$: modality is an access variable, not a free axis. (iii) A second independent
421 temperature-0 run of one model’s 216-cell grid agrees bit-identically on 172/216 cells but shifts the aggregate by
422 +0.073 rad (CI [+0.031, +0.120]), so hosted temperature-0 inference is not exactly repeatable; this measured
423 drift is why hosted results are reproducible only from the saved records.

424 B Training and compute

425 All primary learned models use a 256-wide, 3-layer tanh MLP, Adam, and a cosine-annealed learning rate,
426 and train on a single 8 GB consumer GPU. Mixed- k rollout adds negligible parameters; its cost is wall-clock
427 time (the HNN mixed-window run is the most expensive at ~ 6.7 h because each unroll step recomputes ∇H
428 via autograd). Chronos-2 runs at inference only. The dataset has 67,491 samples across $k \in \{1, 2, 3\}$ and
429 three regimes, subsampled at $\Delta t = 0.01$ s from the 10^{-3} s ground truth. Despite the sample count, the training
430 set contains only *nine* trajectories, one for each (k , regime) cell. The budget study uses nested subsets of 1,
431 5, and 20 trajectories per eligible cell and three training initializations. Neural ODE therefore sees 9, 45, or
432 180 trajectories; HNN sees 3, 15, or 60 undamped trajectories. Evaluation uses the same two held-out seeds
433 throughout.

Table 4: In-sample versus held-out error (rad) for the six stable learned models: mean over the nine training trajectories versus the shared held-out grid, both at horizons $\{1, 10\}$ s with the harness inference path. The in-sample best is the most overfit.

Model	In-sample	Held-out	Ratio
neural-ode	0.625	1.110	1.8
neural-ode-rollout	0.382	1.121	2.9
neural-ode-rollout-mixed	0.092	1.036	11.3
hnn	0.568	1.138	2.0
hnn-rollout	0.560	0.972	1.7
hnn-rollout-mixed	0.522	1.066	2.0

434 Compute cost. The API LLM evaluation is the only billed run (\$110.38 for 3,960 forecasts, June 2026
 435 deployments); local methods cost only compute.

436 Reproducibility records. The anonymized artifact accompanying this submission contains the code,
 437 prompts, trained weights, held-out specifications, and the per-cell records for the main held-out comparison
 438 (2,160 records), the matched disclosure test (1,800), the modality and prompting grid, and the Lorenz sweep;
 439 the two large sweeps, the training-budget and balanced-seed studies, are packaged as separate archives. All
 440 dataset and held-out seeds are fixed and recorded in the released configuration. LLMs are sampled once at
 441 temperature 0. The reported hosted-model numbers are recomputed from the saved records; re-calling the API is
 442 a new evaluation rather than a reproduction. Invalid cells are scored at $\pi/2$.

443 C Seed and system replication

444 The Lorenz-63 test uses $(\sigma, \rho, \beta) = (10, 28, 8/3)$, RK4 ground truth at $\Delta t=0.01$ s, a 5 s context, one fixed
 445 training seed, and five fixed held-out seeds (all recorded in the released configuration). Each held-out seed
 446 contains five trajectories, giving 25 held-out trajectories in total. The Neural ODE is a two-hidden-layer, width-
 447 128 tanh network trained for 300 epochs by analytic derivative matching. SINDy uses a degree-two polynomial
 448 library with STLSQ, fit to analytic derivatives at the sampled training states; the refit selects the true sparsity
 449 pattern exactly and recovers $(\hat{\sigma}, \hat{\rho}, \hat{\beta}) = (10, 28, 8/3)$ with maximum absolute coefficient error 2.2×10^{-13} .
 450 NVAR uses two delays, degree-two features, and ridge regression. Chronos-2 uses `amazon/chronos-2` in
 451 univariate and multivariate modes.

452 D Breakdowns by link count and regime

453 Reliability-adjusted mean angle error (rad) on the shared held-out grid (horizons $\{1, 10\}$ s), by number of links
 454 (Table 6) and by regime (Table 7); models in the order of Table 3. Cells a model never produced (e.g. the LNN's
 455 diverged $k=3$) are scored at $\pi/2$. Each row uses the same grid as Table 3.

456 E Additional result tables

457 Full per-model values for the horizon breakdown (§5.6) and the predictability horizon (§5.2), together with the
 458 regime definitions, the matched disclosure gap, and the training-budget, training-objective, and physical-fidelity
 459 studies. The corresponding summary statistics appear in the main text.

460 F Real-data check details

461 The hardware dataset [Myers et al., 2020] is openly released via Mendeley Data (CC BY 4.0 listing; the deposit
 462 includes a GPLv3 license file) with video tracking of both link angles at 500 Hz and per-frame uncertainty
 463 estimates (median 0.20° upper link, 0.43° lower). We use the published simplified point-mass parameters
 464 ($L = [0.172, 0.143]$ m, $m = [0.311, 0.111]$ kg, $g = 9.81$ m/s²). Angular velocities come from a Savitzky-
 465 Golay filter (window 42 ms, cubic, analytic derivative). The recordings begin with the pendulum positioned
 466 by hand, so the free drop is detected as the first time either link exceeds 4 rad/s; evaluation states are drawn
 467 evenly from 0.2 to 24 s after the drop, 60 per trial. The tracker zeroes its angle at the first frame, leaving a
 468 gravity-referenced offset per link; every trial ends hanging at rest with tracked angles at exact multiples of 360°
 469 (within 0.08°), which fixes the offset at zero independently of any model. A 20 ms physics-tracking check agrees,
 470 with error of 0.005–0.007 rad against a 0.006 rad tracking floor. A grid search over $\{0, 5 \times 10^{-4}, \dots, 2 \times 10^{-2}\}$
 471 selects damping 10^{-3} on Trial 1 at the 0.5 s horizon. SINDy uses a degree-two polynomial plus first-order

Table 5: Replication results. Top: reliability-adjusted pendulum error (rad) by held-out seed and their seed mean; horizons $\{1, 10\}$ s. The primary seed has 20 trajectories per (k, regime) cell and the new seed has five. Equal-seed answer rates are 97% for LNN, 61% for SINDy, and 46% for NVAR; all other rows answer 100%. Bottom: Lorenz-63 normalized MAE, averaged within each of five held-out seeds (five trajectories each) and then across seeds, with the standard error over seeds. SINDy is below 2×10^{-5} in every cell. Lower is better.

Pendulum model	Primary seed	Second seed	Seed mean
rk4 (reference)	0.059	0.074	0.067
symplectic	0.205	0.194	0.200
euler	0.405	0.415	0.410
chronos2-uni	0.697	0.604	0.650
chronos2-multi	0.684	0.621	0.653
sindy	0.860	0.906	0.883
hnn-rollout	0.972	0.858	0.915
neural-ode-rollout-mixed	1.036	0.989	1.012
hnn-rollout-mixed	1.066	1.037	1.051
nvar	1.245	1.251	1.248
lnn	1.356	1.376	1.366
Lorenz model	0.1 s	0.5 s	1.0 s
rk4 (reference)	0.000	0.000	0.000
sindy	$<10^{-5}$	$<10^{-5}$	$<2 \times 10^{-5}$
neural-ode	0.014 (0.008)	0.079 (0.030)	0.277 (0.063)
nvar	0.012 (0.001)	0.262 (0.050)	0.429 (0.084)
chronos2-uni	0.126 (0.027)	0.700 (0.102)	0.732 (0.085)
chronos2-multi	0.101 (0.018)	0.691 (0.129)	0.771 (0.097)
euler	0.028 (0.005)	0.216 (0.023)	0.885 (0.120)
persistence	0.524 (0.054)	1.028 (0.123)	0.900 (0.057)

Table 6: Out-of-sample error by number of links k .

Model	$k=1$	$k=2$	$k=3$
rk4 (reference)	0.000	0.068	0.109
symplectic	0.000	0.253	0.363
euler	0.271	0.410	0.534
kimi-k2.6	0.112	0.553	0.949
chronos2-multi	0.263	0.798	0.992
chronos2-uni	0.279	0.813	0.999
grok-4-1-fast-reasoning	0.366	1.030	1.247
deepseek-v4-pro	0.872	0.949	1.036
hnn-rollout	0.524	0.985	1.406
neural-ode-rollout-mixed	0.845	1.178	1.084
hnn-rollout-mixed	0.550	1.162	1.487
neural-ode	0.626	1.304	1.400
neural-ode-rollout	0.968	1.172	1.224
hnn	0.511	1.337	1.567
lnn	1.072	1.440	1.556

Table 7: Out-of-sample error by regime. Learned-model hidden cells use the equalized hidden-regime run. The hidden regime remains easier for most models because its low-gravity constants give slower, less-divergent motion. The unadjusted regime contrast in §5.2 consequently mixes disclosure with difficulty.

Model	normal	disclosed	hidden
rk4 (reference)	0.174	0.003	0.000
symplectic	0.387	0.177	0.051
euler	0.743	0.387	0.085
kimi-k2.6	0.635	0.519	0.461
chronos2-multi	0.818	0.826	0.409
chronos2-uni	0.818	0.798	0.475
grok-4-1-fast-reasoning	1.096	0.959	0.588
deepseek-v4-pro	1.240	1.097	0.520
hnn-rollout	1.087	1.115	0.713
neural-ode-rollout-mixed	1.260	0.836	1.011
hnn-rollout-mixed	1.244	1.172	0.782
neural-ode	1.221	1.095	1.014
neural-ode-rollout	1.281	0.894	1.189
hnn	1.262	1.303	0.850
lnn	1.525	1.416	1.127

Table 8: Matched disclosure gap (rad), paired on identical trajectories. $\Delta_{\text{disc}} = \text{err}(\text{hidden}) - \text{err}(\text{disclosed})$; positive values indicate higher error without the constants. Reliability-adjusted; paired Wilcoxon, Holm-corrected. This experiment uses its own matched grid (3 models $\times$ 2 horizons $\times$ 150 trajectories = 900 pairs), not the shared held-out set, so its absolute errors are not comparable to those in Table 7. Per-model Δ values are computed before rounding.

Model	Disclosed	Hidden	Δ_{disc}
kimi-k2.6	0.746	0.957	+0.211
grok-4-1-fast-reasoning	0.987	1.025	+0.038
deepseek-v4-pro	1.425	1.382	−0.042
pooled (paired, $n=900$)			+0.069*
$k=1$			+0.297***
$k=3$			−0.055*

* $p_{\text{Holm}} < 0.05$, *** $p_{\text{Holm}} < 10^{-3}$.

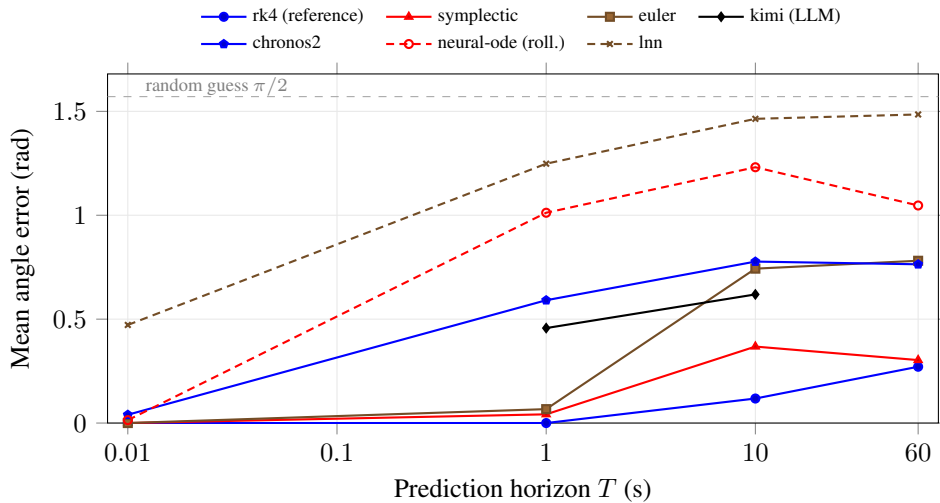


Figure 1: Error growth with prediction horizon (out-of-sample, reliability-adjusted; log time axis). The numerical integrators stay near zero until ~ 10 s; every black-box family saturates near the chaotic noise floor by 1 s. LLMs are evaluated at $\{1, 10\}$ s only. Dashed line: random-guess baseline $\pi/2$.

Table 9: Physical regimes. The changed_hidden regime withholds the constants.

Regime	g (m/s ²)	L (m)	m (kg)	damping	disclosed
normal	9.81	1.0	1.0	0.0	—
changed_disclosed	3.71	1.4/0.7/1.1	2.0/0.6/1.5	0.05	yes
changed_hidden	1.62	0.8/1.2/0.9	0.5/1.8/1.0	0.10	no

Table 10: Nested training-budget study. Reliability-adjusted pendulum error (rad), averaged over three training initializations and two held-out seeds. Intervals come from hierarchical resampling over initializations and held-out seeds. HNN is evaluated on the undamped normal regime only.

Model	Scope	Total trajectories	Error	95% CI
HNN	normal	3	1.269	[1.209, 1.349]
HNN	normal	15	0.974	[0.891, 1.062]
HNN	normal	60	0.981	[0.856, 1.113]
Neural ODE	all regimes	9	1.090	[1.011, 1.207]
Neural ODE	all regimes	45	0.968	[0.923, 1.028]
Neural ODE	all regimes	180	0.861	[0.829, 0.907]

Table 11: Training-objective study, out-of-sample mean angle error (rad) on the shared held-out grid (horizons $\{1, 10\}$ s); architecture fixed, loss/window varied. Significance from a powered paired study on the full four-horizon held-out set ($n=720$, Wilcoxon+Holm).

Model	Objective	Err	vs. prev. row
Neural ODE	derivative MSE	1.110	—
	rollout (single- k)	1.121	n.s. ($p_{\text{Holm}}=0.188$)
	mixed-window	1.036	sig. ($p_{\text{Holm}}=0.013$)
HNN	derivative MSE	1.138	—
	rollout (single- k)	0.972	sig. ($p_{\text{Holm}}=1.95 \times 10^{-4}$)
	mixed-window	1.066	sig. higher ($p_{\text{Holm}}=0.014$)

Table 12: Physical fidelity on the undamped (normal) regime: mean absolute energy drift over a 10 s rollout, normalized by $E^*=g \sum_i m_i L_i$ (0 = perfect conservation), and fraction of bounded rollouts. HNN drift is nearly unchanged when its rollout uses a symplectic integrator.

Model	Family	Energy drift	Stable
rk4	numerical	0.000	100%
symplectic	numerical	0.066	100%
euler	numerical	0.494	100%
neural-ode-rollout-mixed	learned	0.779	100%
hnn-rollout-mixed	learned	0.865	100%
neural-ode-rollout	learned	0.872	100%
hnn-rollout	learned	2.065	100%
lnn	learned	4.038	27%
hnn	learned	4.270	100%
neural-ode	learned	4.346	100%

Table 13: Out-of-sample mean angle error (rad) by horizon, reliability-adjusted, on the shared held-out grid (selected models). The cross-family comparison uses horizons $\{1, 10\}$ s.

Model	1 s	10 s
rk4 (reference)	0.000	0.118
symplectic	0.042	0.368
euler	0.067	0.743
kimi-k2.6	0.457	0.619
chronos2-multi	0.591	0.777
chronos2-uni	0.622	0.773
grok-4-1-fast-reasoning	0.905	0.857
deepseek-v4-pro	0.996	0.909
hnn-rollout	0.855	1.088
neural-ode-rollout-mixed	0.951	1.120
hnn-rollout-mixed	0.982	1.150
neural-ode	0.892	1.328
neural-ode-rollout	1.012	1.231
hnn	1.052	1.224
lnn	1.248	1.464

Table 14: Predictability horizon (s): longest horizon tracked contiguously below threshold τ , on the shared out-of-sample set (ladder $\{0.01, 1, 10, 60\}$ s). Higher is better; “—” means not even the first horizon stays below τ .

Model	$\tau=0.1$	0.25	0.5	1.0
rk4 (reference)	1.0	10	60	60
symplectic	1.0	1.0	60	60
euler	1.0	1.0	1.0	60
neural-ode-rollout	0.01	0.01	0.01	0.01
neural-ode-rollout-mixed	0.01	0.01	0.01	1.0
chronos2-multi	0.01	0.01	0.01	60
chronos2-uni	0.01	0.01	0.01	60
hnn-rollout	0.01	0.01	0.01	1.0
neural-ode	0.01	0.01	0.01	1.0
hnn	0.01	0.01	0.01	0.01
lnn	—	—	0.01	0.01

472 Fourier library on Trial 1 with Savitzky–Golay derivatives; Chronos-2 receives a 5 s context resampled to 50 Hz
473 and forecasts at most 50 steps. Failures are scored at $\pi/2$; Chronos-2’s are input-infeasible cells (insufficient
474 context) rather than divergences, and the fixed-coverage rule scores both alike. The tracking dataset has been
475 public since 2020 and may appear in foundation-model corpora; contamination would inflate Chronos-2’s score,
476 so its baseline-level result is, if anything, conservative. Paired bootstrap differences over the 120 states at
477 1.0 s: RK4 minus persistence $+0.018$ (95% CI $[-0.121, +0.155]$); damped minus undamped RK4 -0.103
478 ($[-0.202, -0.003]$), so the damping gain at the longest horizon is significant while the RK4–persistence gap
479 is not. States sampled along a trajectory are temporally dependent, so we repeat both with a moving-block
480 bootstrap within each trial (blocks of 10 consecutive states, ~ 4 s): the conclusions are unchanged (damped
481 minus RK4 $[-0.179, -0.019]$; RK4 minus persistence $[-0.131, +0.064]$), and remain so for block lengths
482 from 2 to 12 s.

Table 15: Real-data check: mean wrapped angle error (rad) over both links on Trials 2–3 (120 held-out states), failures at $\pi/2$. The physics rows use the published constants; “damped” adds the Trial-1-fitted viscous coefficient. Tracking noise floor ~ 0.006 rad.

Model	0.1 s	0.5 s	1.0 s	Answered
physics-rk4	0.084	0.820	1.248	100%
physics-rk4-damped	0.086	0.892	1.145	100%
nearest-neighbor (cross-trial)	0.371	0.849	1.117	100%
chronos2	0.611	1.160	1.223	89%
constant-velocity	0.345	1.402	1.576	100%
sindy-real	0.490	1.501	1.535	99%
persistence	0.934	1.654	1.230	100%